

Platform Team

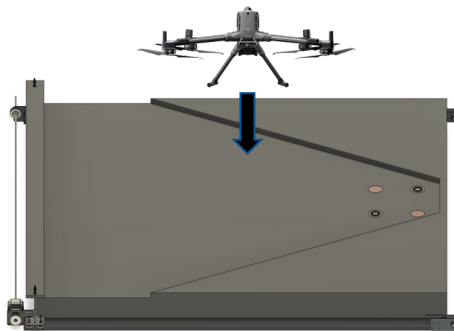
End-to-End Platform Operational Workflow

Platform enables soft landing, minimizes impact, and secures the drone at a centralized location

1

Manual Approach and Landing

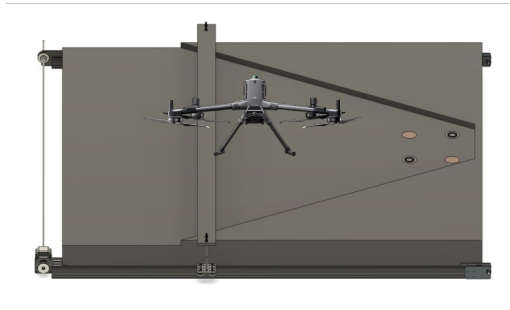
Drone lands in the open space of the platform



2

Passive Alignment and Centering

Mechanisms guide the drone to a precise, centralized position



3

Complete Docking

Drone is...

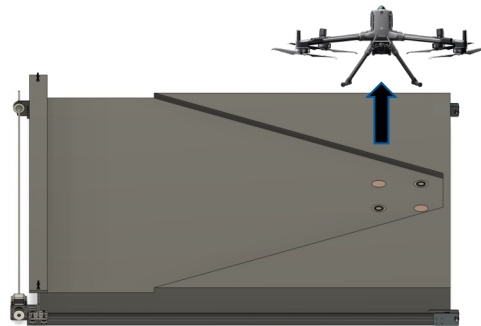
- Positioned
- Magnetized
- Charged



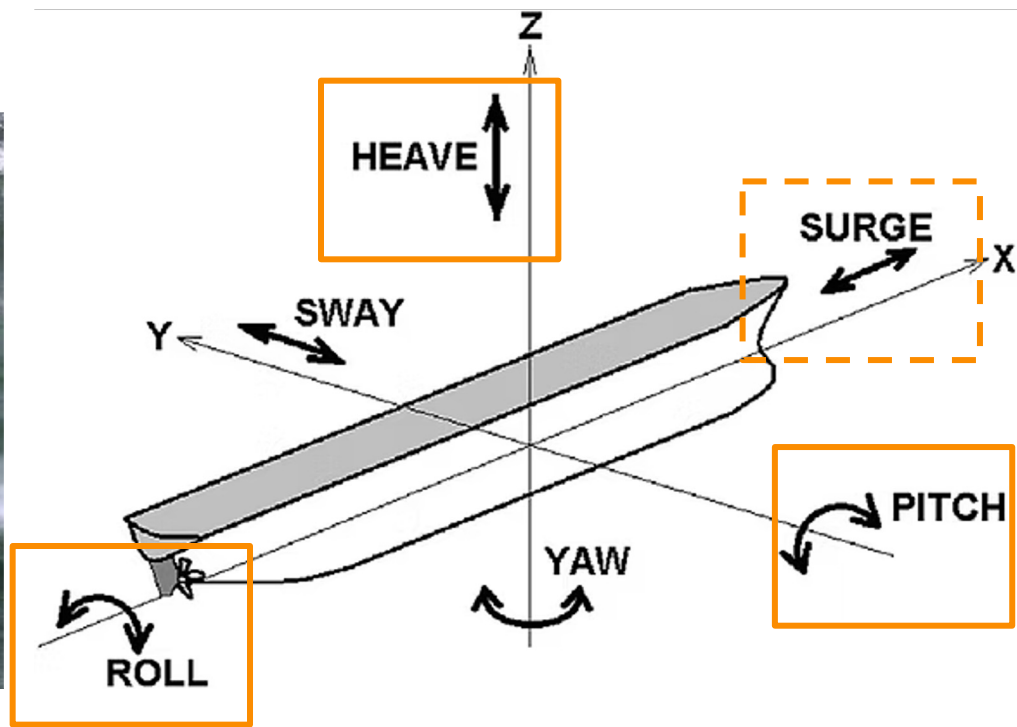
4

Relaunch

- Clamping bar releases
- Magnets deactivates
- Charging deactivates
- Drone relaunches



Boat Motions



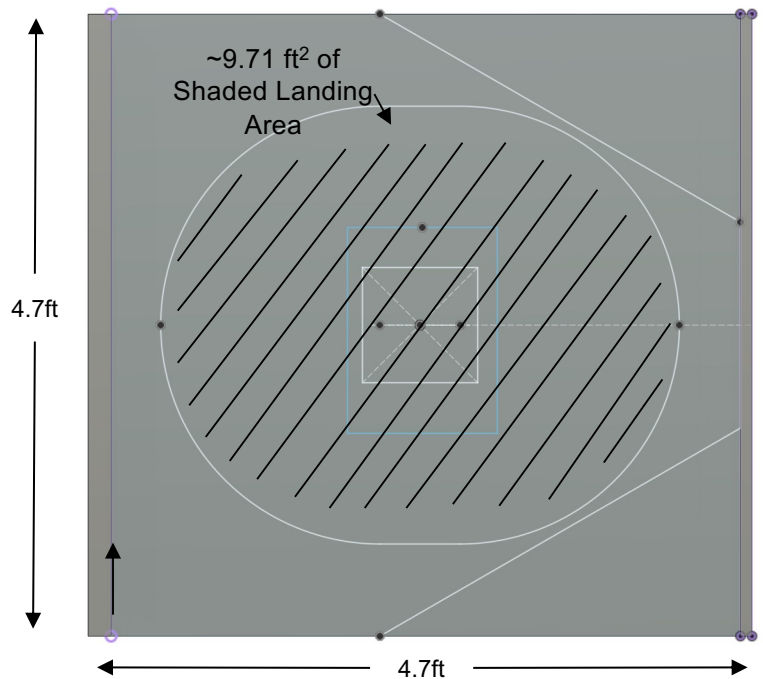
Boat Motions

Quantity	Maximum experienced values	Why we care
Pitch	14° 20° /s	Drones cannot takeoff/land in >10 degrees
Roll	17° 20° /s	
Heave	1 m/s	
Surge	0.4 m/s	Adds landing error to drone

Derived from combination of MAXSurf simulations & NOAA Buoy Data

Expected Drone Landing Area & Allowable Deviation

Identified landing deviation to determine required platform dimensions



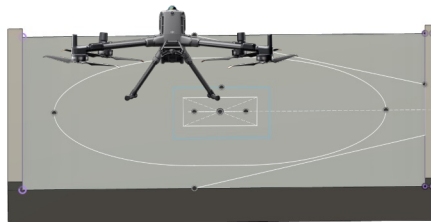
Shaded Region
Indicates Drone
Landing Area



Non-Shaded Region
Indicates Factor of
Safety for Landing

Key Drone Landing Assumptions:

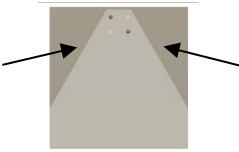
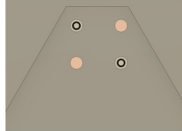
- Landing position within a 20" radial allowance
- No lateral motion only surge motion of the boat



Boat motion and waves introduce
variability in landing

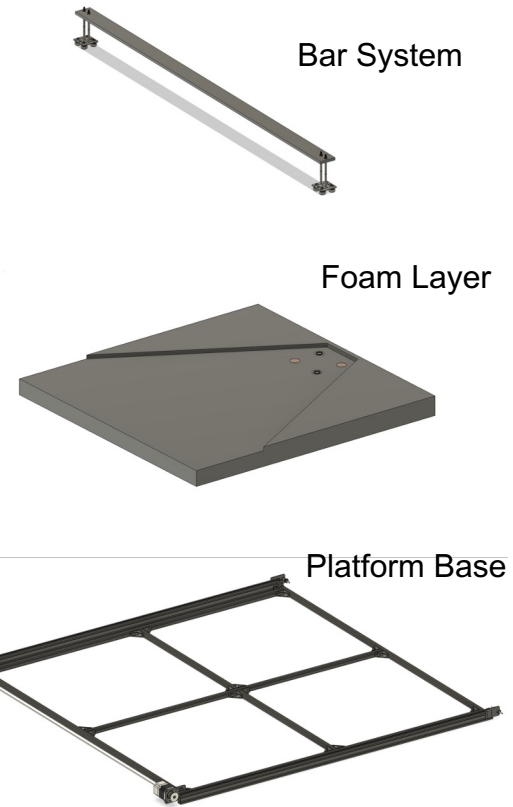
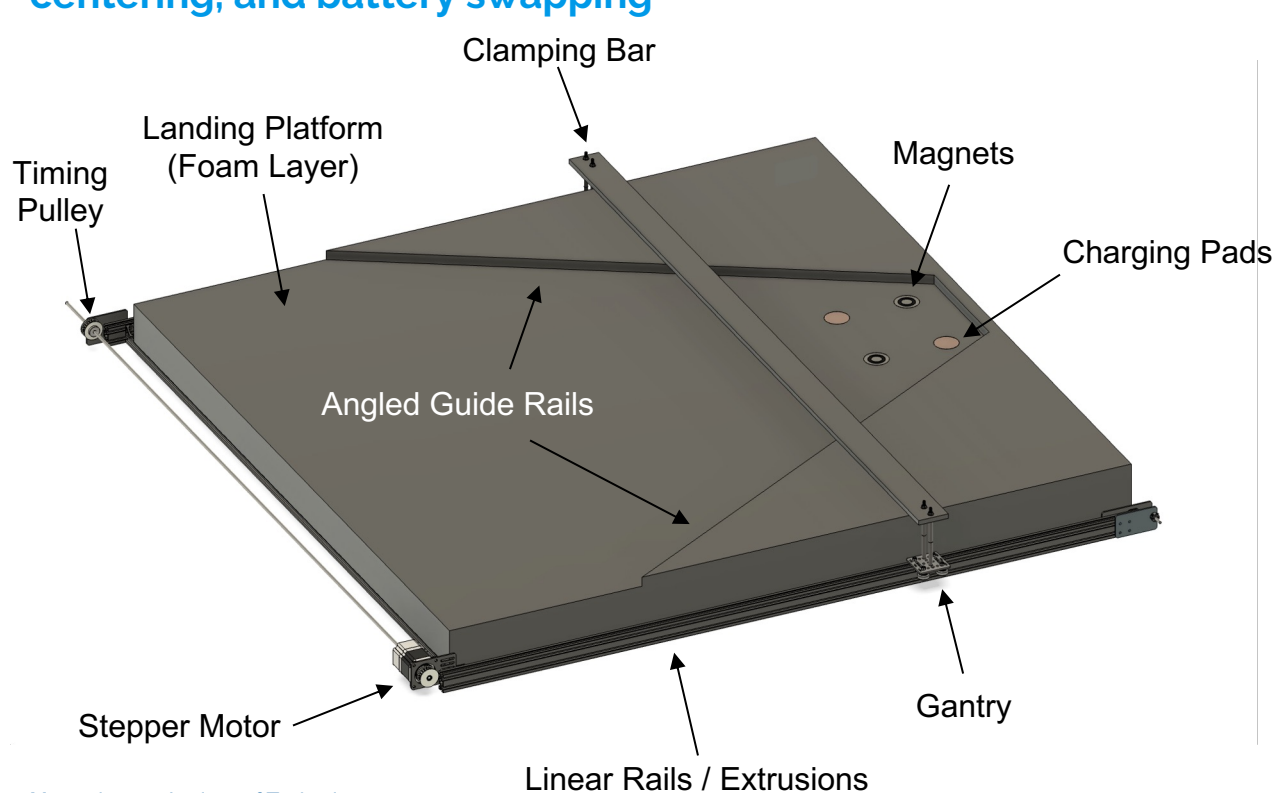
Platform Design Mapping

Each design choice and platform subsystem is tied to a specific operational constraint

High-Level Challenge	Design Requirement	Our Solution	Design
Hard landings risk drone sensor damage	Soft, impact-reducing landing surface	5x5 feet Foam layer for impact damping	
Drone must move to platform edge for battery swapping capability	Final position along platform edge	Gantry rail system with drone on top of vinyl layer for smooth motion	
Drone may land in varying positions/orientations	Consistent, repeatable positioning and orientation	Angled guides to align and orient drone repeatably	
Drone faces wave dynamics and accelerations	Securement to prevent moving in final position	Embedded magnets in foam layer	

Platform CAD: System Architecture & Assembly

Demonstrates how integrated components work together to achieve precise clamping, centering, and battery swapping



Passive Centering via Angled Guide Rails

Angle bars and roller bearings guide the drone to a fixed swap position

Drone Boat - Stabilization Control

Dashboard OPS Console

LIVE VIDEO ws://100.127.225.126:8765

Drone live

FLIGHT MODE
AUTO.LOITER

BATTERY
16.32 V 100.0 %

STATUS
Armed No
FCU connected Yes

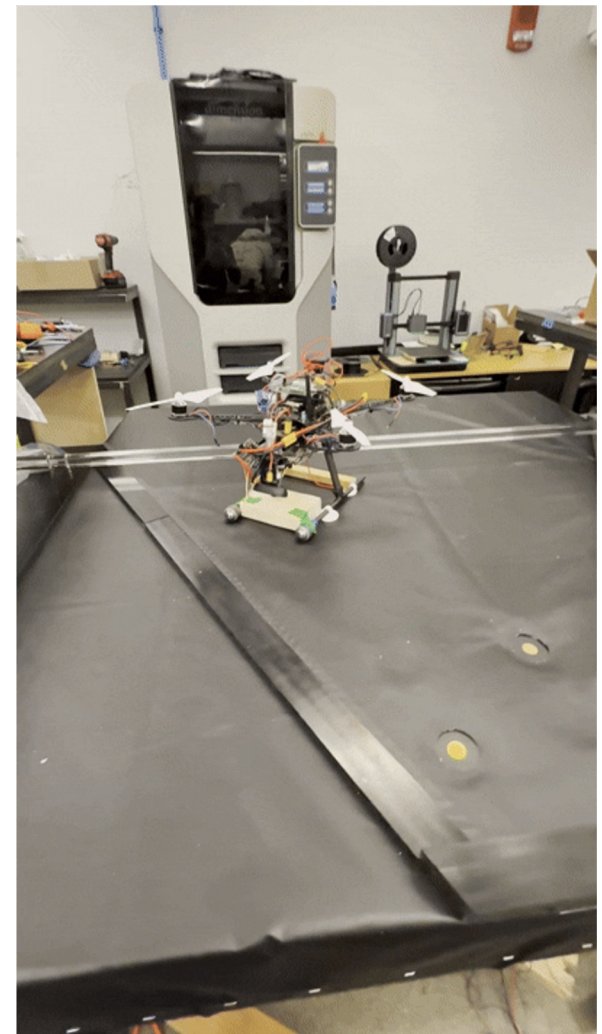
PLATFORM /dev/ttyUSB1 - 115200
Forward (Slower) <- Backward (Faster) >- Stop <->
Magnet On <- Magnet Off >- Contact On <->
Contact Off >-

HOUSING /dev/ttyUSB0 - 115200
Open <- Close >-

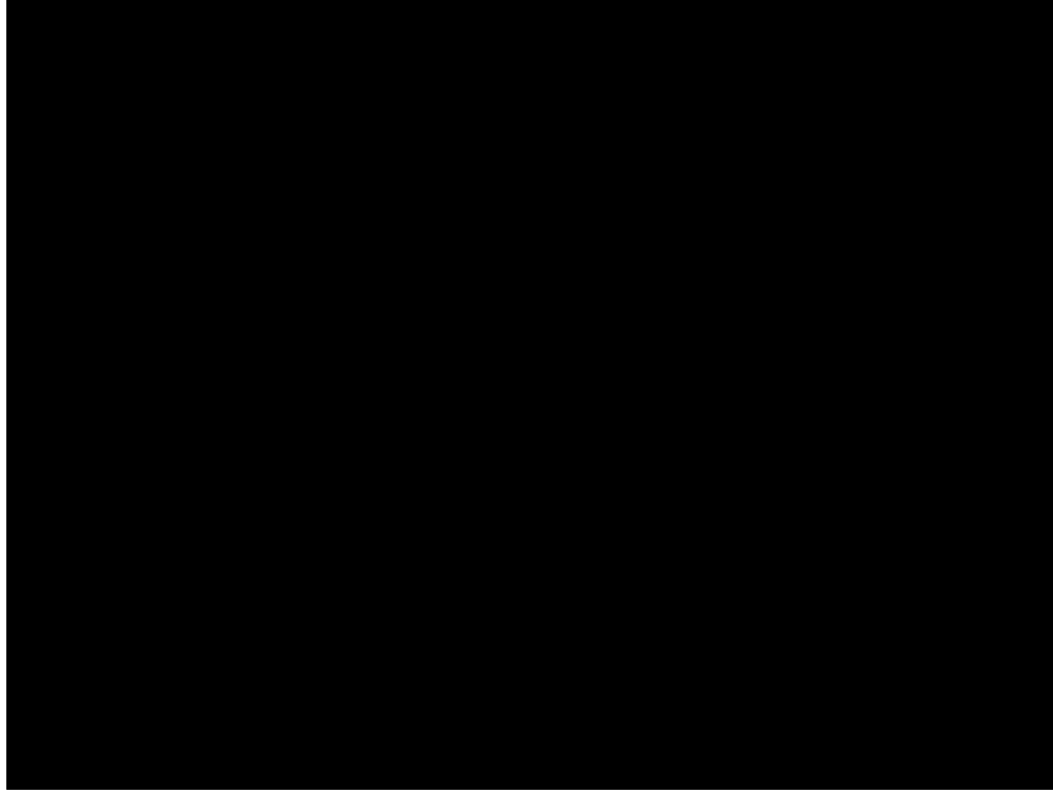
STABILIZATION /dev/ttyUSB2 - 115200
read-only: no commands

Available Commands

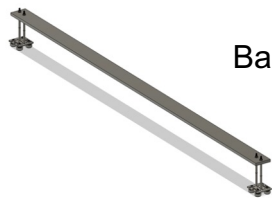
Charging Indicator



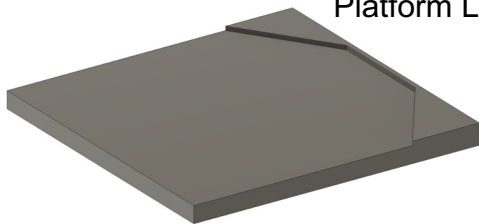
Contact Based Charging for Drone Platforms



Platform Bill of Materials



Bar System



Platform Layer



Platform Base

Subsystem	Total Subassembly
Bar System	\$290.72
Platform Layer	\$300.00
Platform Base	\$1211.64
TOTAL	\$1802.36